CHAPTER 3

DISCUSSION OF FINDINGS

This chapter discusses the findings, identified workforce-management challenges, implemented WorkPulse modules, respondent evaluations, and the technical evidence required to substantiate system performance. Survey results describe respondents' perceptions; functional, security, reliability, offline, and biometric claims require separately documented technical tests.

Challenges and Issues of the Current System

The existing workforce processes were reported to involve partly manual or fragmented attendance, employee information, leave, and payroll records. The frequencies below are retained from the submitted Chapter 3 and use the study population of 30 respondents.

Challenge	Frequency	Percentage
Inaccurate attendance recording	27	90.00%
Risk of unauthorized or proxy attendance	28	93.33%
Inefficient monitoring of work hours	21	70.00%
Payroll processing difficulties	27	90.00%
Limited employee access to records	28	93.33%

The most frequently identified concerns were proxy attendance and limited employee access to records (28 of 30 respondents each). Attendance inaccuracy and payroll difficulties were each identified by 27 respondents, while 21 reported difficulties monitoring work hours. These findings describe the respondents' reported challenges rather than independently measured error rates.

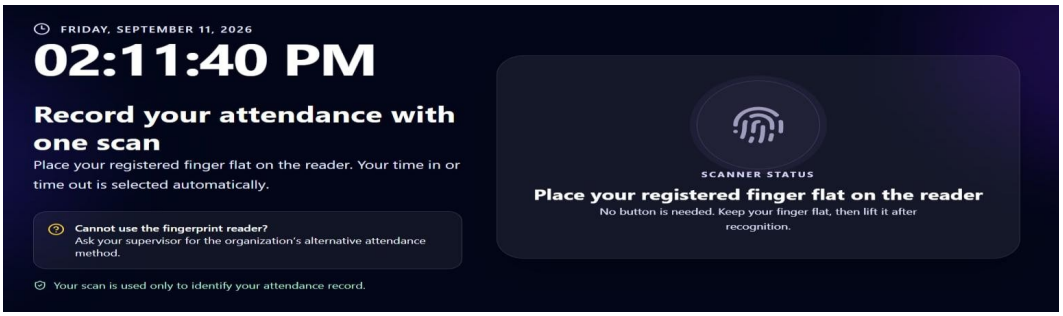
Implemented Features of WorkPulse

The following descriptions reflect the revised system scope. The final deployed version and its screenshots must be checked before each feature is described as verified by testing.

Employee Information Management

Authorized administrators manage centralized employee records. The employee classifications are Regular, Extra, and Manager / Supervisor. These classifications describe workforce categories and do not create separate application access groups. Any older screenshot showing Manager and Supervisor as separate options must be replaced with the verified combined classification.

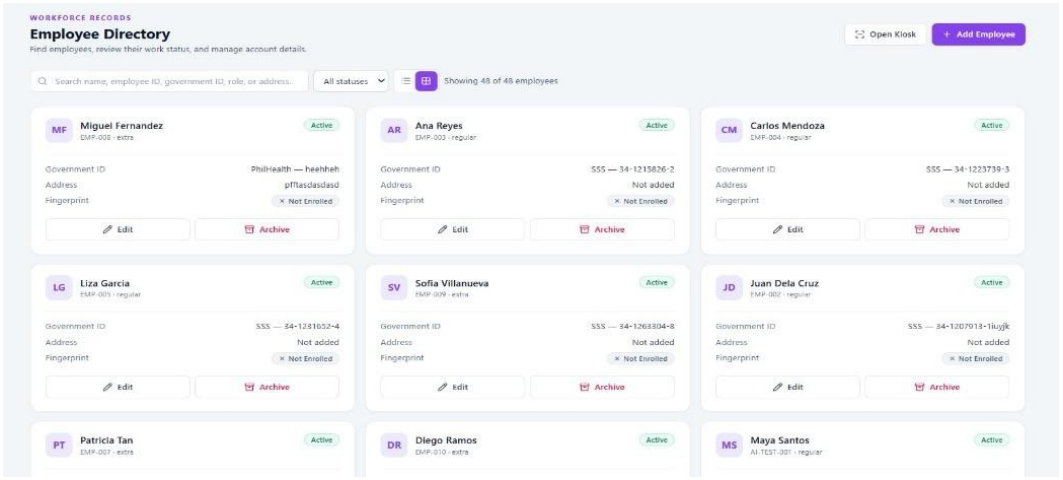
Figure 3. Employee Information Management (original manuscript image; replace if outdated).



Fingerprint-Based Attendance

The attendance kiosk supports fingerprint-based time-in and time-out through a configured local scanner workstation, reader drivers/WebSDK, and local fingerprint capture and matching components. The hosted Render backend does not directly access a USB scanner attached to a user's computer. Fingerprint operation requires the supported workstation environment, scanner software, and connectivity to the relevant online attendance services. The exact tested operating system, device model, matching-software version, network conditions, and trial results must be recorded in the biometric evaluation subsection.

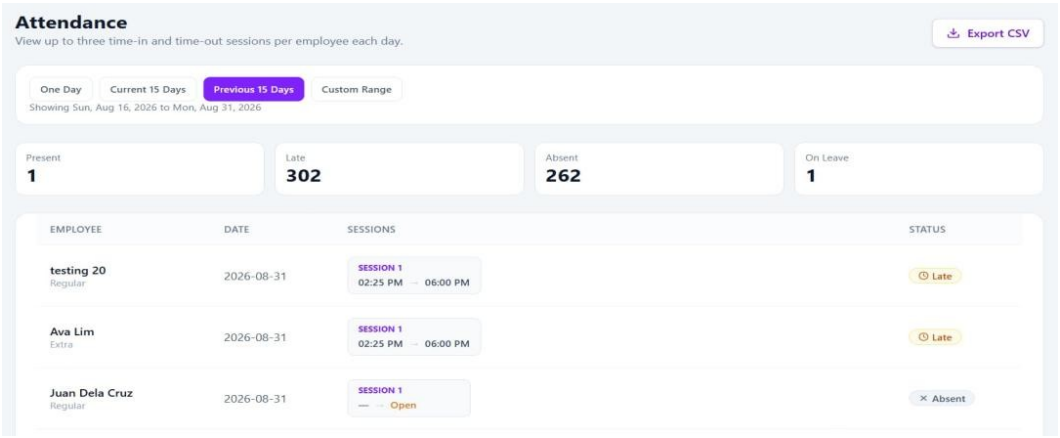
Figure 4. Fingerprint-Based Attendance (original manuscript image; replace if outdated).



Attendance Monitoring and Corrections

Administrators can view daily attendance and filter by date range. Authorized administrators can manually record attendance or correct eligible records, with a required reason, actor and timestamp history, and an employee notification. Attendance displays distinguish recorded and edited entries and identify their source. Attendance records included in settled payroll must not be silently edited; the system should reject prohibited changes and preserve the audit trail. These behaviors require final deployment verification.

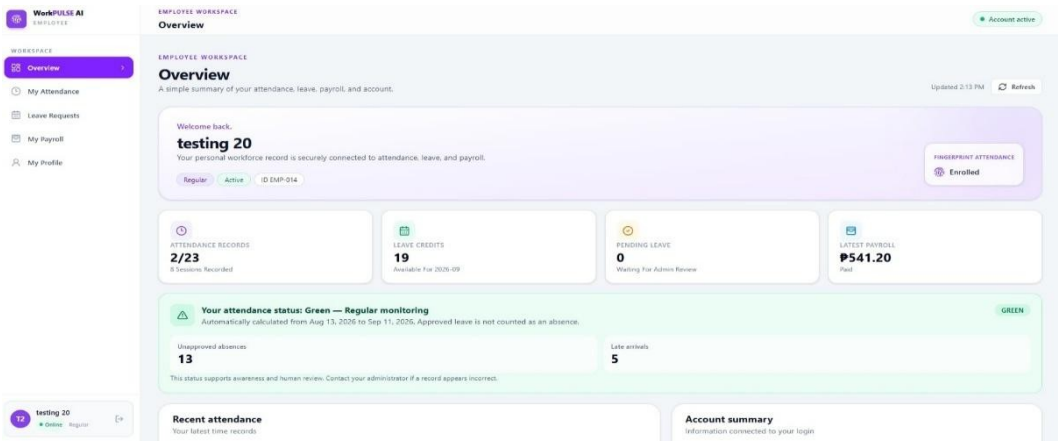
Figure 5. Attendance Monitoring (original manuscript image; replace if outdated).



Employee Self-Service Portal

Employees can view authorized personal records, attendance, leave, payroll and payslip information, and the status of cash advance requests. The notification bell surfaces relevant attendance corrections and other employee notices. Attendance entries indicate whether they were fingerprint-recorded or manually recorded/edited. Limited offline operation permits selected previously cached views and queues eligible actions, with clear warnings that cached information may be outdated. Authorized administrators, rather than a separate HR role, manage the relevant approvals and corrections.

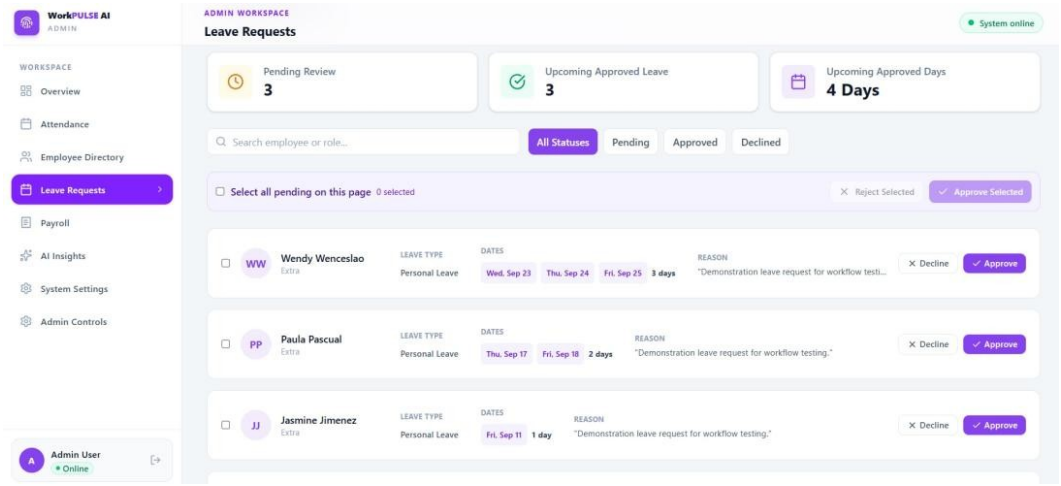
Figure 6. Employee Self-Service Portal (original manuscript image; replace if outdated).



Leave Management

Employees submit leave requests and view their status. Authorized administrators review and act on requests. Eligible requests may be saved to a local offline queue and submitted after reconnection; an offline queue entry is not an approval.

Figure 7. Leave Management (original manuscript image; replace if outdated).



Payroll Management and Payslip Summary

The payroll workflow distinguishes Ready to Pay, Carried to Next Period, and Paid. Ready to Pay identifies an amount prepared for disbursement; Carried to Next Period tracks unpaid or unsettled amounts transferred forward; Paid denotes confirmed settlement. Payroll and payslip summaries present the relevant gross components, deductions, cash advance repayments, carried balances, and net amount. A proposed repayment is not a recorded repayment until the appropriate payroll/payment event is confirmed and posted. Outstanding cash advance balances are reduced only by confirmed repayments.

Figure 8. Payroll Management (original manuscript image; replace if outdated).

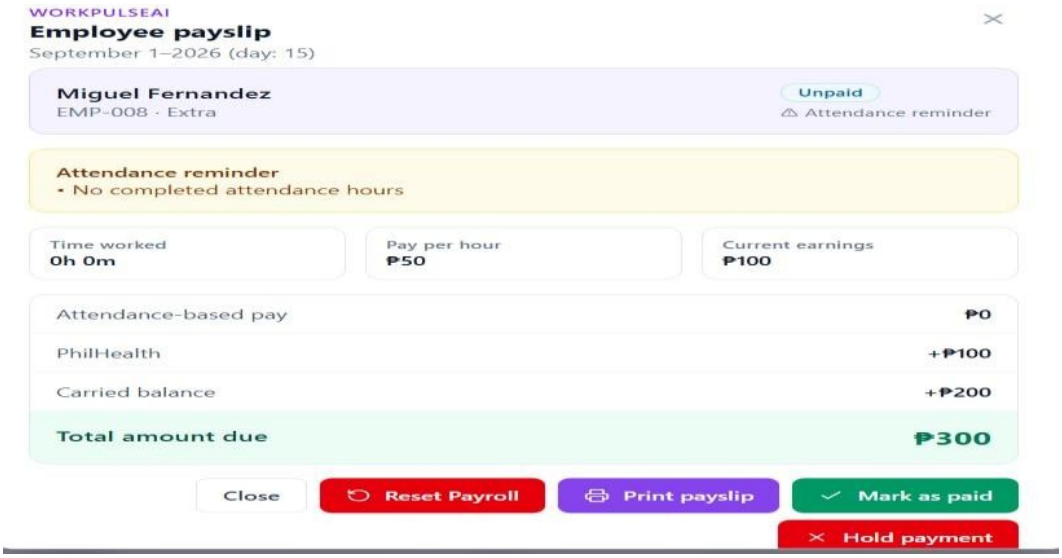
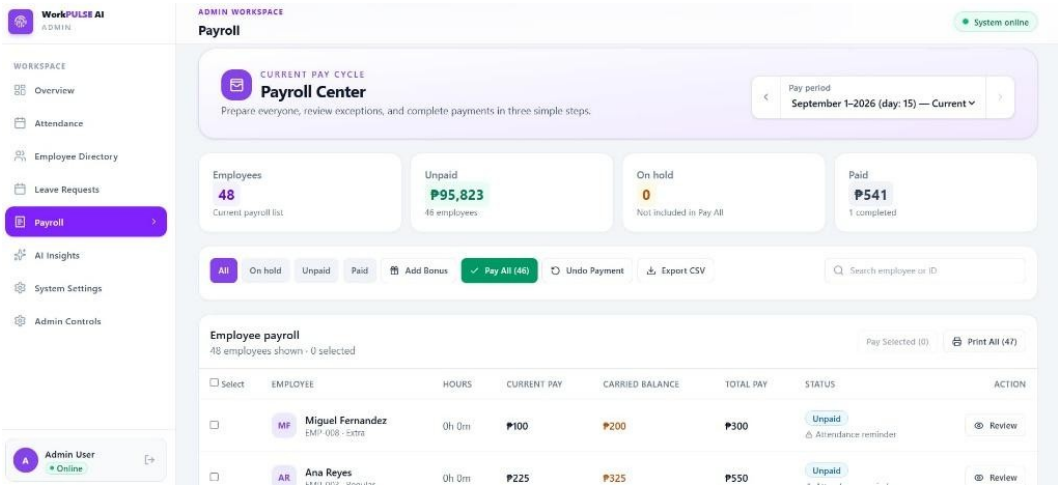


Figure 9. Payroll Summary / Payslip (original manuscript image; replace if outdated).



Cash Advance Management

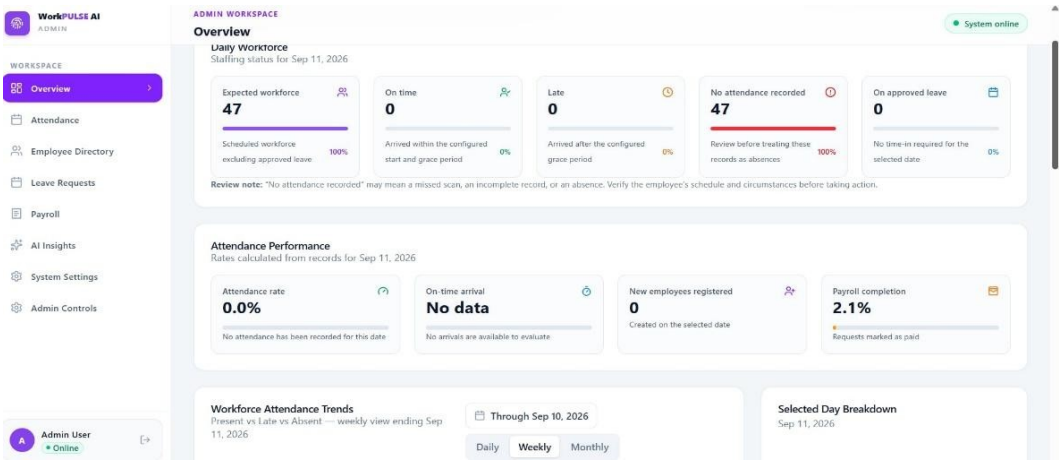
Employees may submit cash advance requests. Authorized administrators review and approve or reject requests; approval is distinct from confirmation that cash was actually given. Once disbursement is confirmed, the system tracks the outstanding amount, repayment entries, and remaining balance. Payroll summaries distinguish scheduled or projected deductions from repayments that have actually been posted.

[INSERT VERIFIED SCREENSHOTS: employee request, administrator decision, disbursement confirmation, balance/repayment history.]

Workforce Analytics Dashboard

Descriptive reporting presents historical counts, rates, tables, and charts. Analytical forecasting, where deployed and evaluated, is a separate function and must be identified by its actual model status. Statistical fallback results must not be described as predictions from a trained AI model. Only implemented and evaluated forecasting functions should be claimed.

Figure 10. Workforce Analytics Dashboard (original manuscript image; replace if outdated).

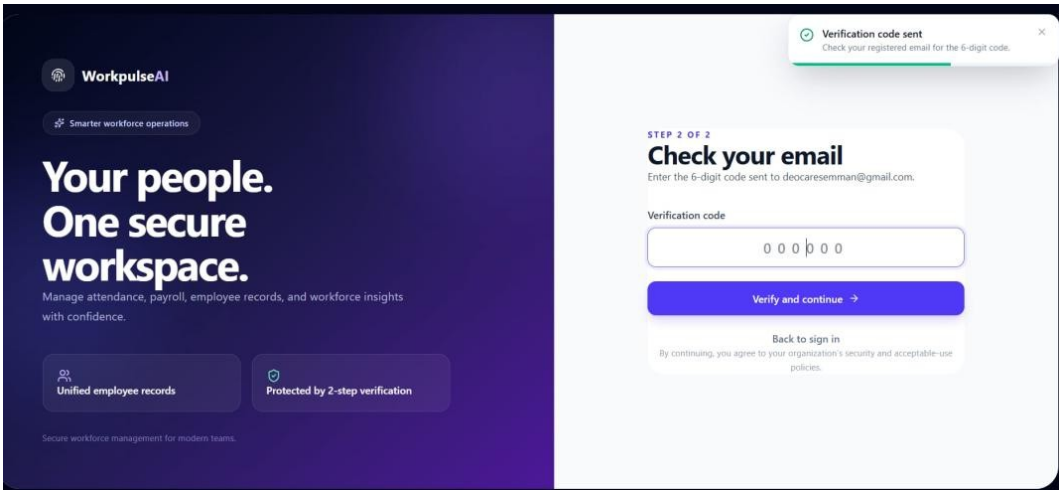


Role-Based Access Control

WorkPulse uses Administrator and Employee access groups. Regular, Extra, and Manager / Supervisor are employee classifications, not separate application roles. Server-side authorization

and record-ownership checks govern access, regardless of which controls are displayed in the browser.

Figure 11. Role-Based Access Control (original manuscript image; replace if outdated).



Secure Authentication and Audit Logging

Authentication includes password and configured email verification. Audit logs record relevant sensitive actions, including attendance corrections, payroll changes and approvals, with sufficient context for accountability.

Figure 12. Secure Authentication (original manuscript image; replace if outdated).

Figure 13. Audit Logging (original manuscript image; replace if outdated).

Data Validation, Security and Cloud Storage

The backend validates requests and enforces role restrictions and record ownership. MongoDB Atlas stores authorized workforce records; browser-local IndexedDB holds limited queued actions and cached views for supported offline behavior, not a complete offline replica of the cloud database.

Figure 14. Data Validation and Security (original manuscript image; replace if outdated).

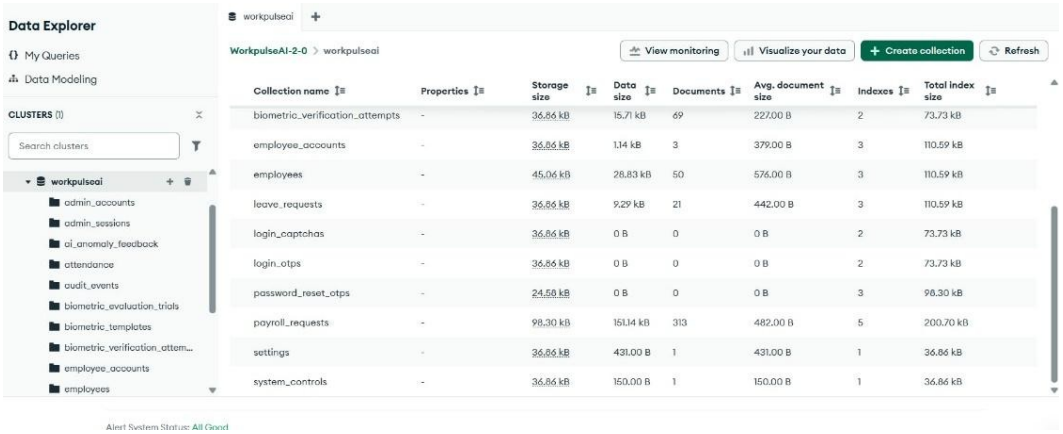
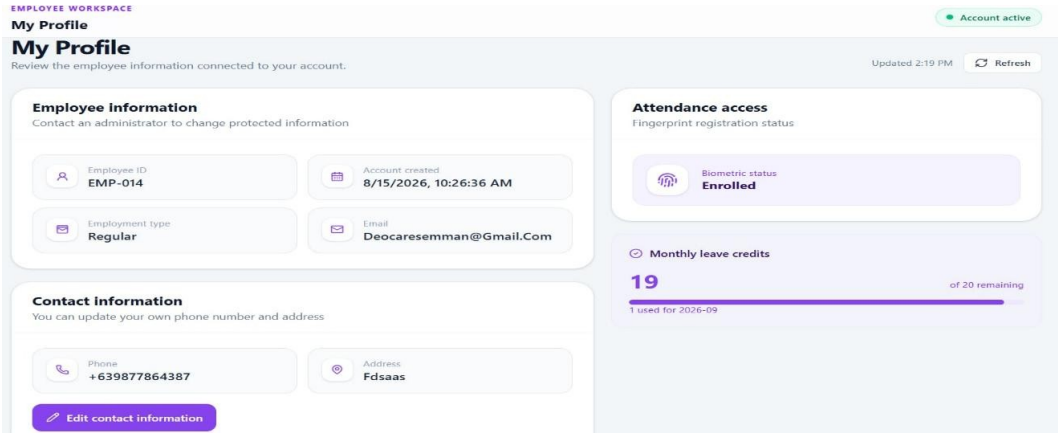


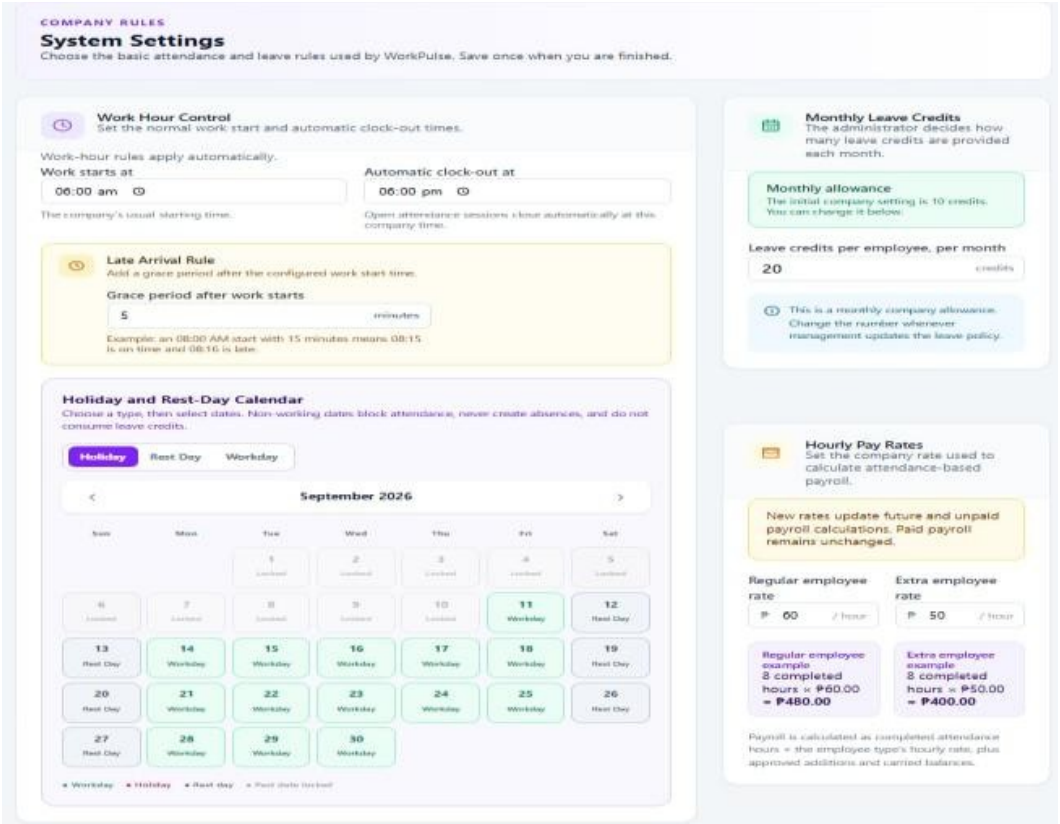
Figure 15. Cloud-Based Data Storage (original manuscript image; replace if outdated).



Company and System Settings

Authorized administrators manage applicable company preferences and the three employee rate classifications: Regular, Extra, and Manager / Supervisor. The final screenshot must show a single combined Manager / Supervisor rate setting.

Figure 16. Company and System Settings (original manuscript image; replace if outdated).



Offline Saving and Synchronization

The limited offline design may queue supported manual attendance/corrections, leave requests, and cash advance requests, and display previously cached information. The interface should distinguish Pending (saved locally but not accepted by the server), Needs Review (server

validation/conflict requires attention), and Synced (accepted by the server). Upon reconnection, the server revalidates identity, authorization, dates, payroll settlement restrictions, and duplicates before accepting each operation. Duplicate-submission prevention and account-specific local storage are required. Fingerprint scanning, payment/disbursement confirmation, approvals, fresh login and uncached data retrieval require connectivity. Network interruptions and local-storage failure must produce understandable errors without falsely reporting success.

[INSERT VERIFIED SCREENSHOTS: Pending, Needs Review, Synced, stale-cache notice, and storage/network error toast.]

Technical Evaluation of Implemented Functions

The following matrix is an evidence checklist, not fabricated test results. Enter the observed outcome, execution date, test ID, and screenshot/log reference from the final deployed system before reporting a test as passed.

Test case	Expected outcome	Observed outcome / evidence
Future attendance time	Reject or flag future-dated attendance	TO VERIFY — add test ID and result
Overlapping sessions	Prevent invalid overlapping sessions	TO VERIFY — add test ID and result
Settled payroll attendance edit	Reject direct modification and preserve history	TO VERIFY — add test ID and result
Unauthorized record access	Deny access to another employee’s private records	TO VERIFY — add test ID and result
Offline queue and reconnection	Queue eligible action; revalidate and synchronize	TO VERIFY — add test ID and result
Duplicate synchronization	Accept a queued operation at most once	TO VERIFY — add test ID and result
Network interruption	Preserve pending operation and report connection issue	TO VERIFY — add test ID and result
Local storage failure	Display error; do not claim local save succeeded	TO VERIFY — add test ID and result
Payroll carried balance	Carry unsettled amount to next period accurately	TO VERIFY — add test ID and result
Cash advance repayment	Post repayment only upon confirmed transaction	TO VERIFY — add test ID and result

Biometric Evaluation Results

A separate biometric testing record is required. Document participant count, enrollment and consent, number of enrolled templates, genuine identification trials, unrecognized/impostor trials using fingers or individuals not enrolled in the evaluation database, failed captures, workstation and scanner conditions, and response-time measurements. For one-to-many identification, an enrolled participant presenting an enrolled finger is a genuine trial, not an impostor trial. Report

false acceptances and false rejections with denominators based on valid comparisons, and list failed captures separately. No numeric biometric accuracy or speed results were supplied in the uploaded chapter, so none are invented here.

Biometric metric	Result	Evidence
Genuine valid trials	TO VERIFY	Trial log
Unrecognized valid trials	TO VERIFY	Trial log
False rejections / genuine valid trials	TO VERIFY	Trial log
False acceptances / unrecognized valid trials	TO VERIFY	Trial log
Failed captures	TO VERIFY	Capture log
Mean/median verification time	TO VERIFY	Timestamp measurements

Usability and Respondent Evaluation

Profile of the Respondents

The respondent classifications were corrected to match the confirmed distribution without changing the total sample size.

Classification	Frequency	Percentage
Owner	2	6.67%
Supervisor / Manager	2	6.67%
Regular Employees	10	33.33%
Extra Employees	16	53.33%
Total	30	100%

Extra employees comprise the largest respondent group (53.33%), followed by regular employees (33.33%). Owners and supervisors/managers each represent 6.67% of the sample.

System Usability Scale (SUS)

The original manuscript reports a SUS score of 80.83, rated Good and Acceptable. This value is retained as a reported result, pending reconciliation with all 30 individual questionnaires and the item-level calculation sheets. Standard SUS scoring converts responses to 0–4 contributions (odd items: response minus 1; even items: 5 minus response), sums contributions and multiplies by 2.5 for each respondent; the overall SUS score is the mean of respondent scores. Include the instrument, individual scoring and summary worksheet in the appendices.

Measure	Reported value
SUS Score	80.83 (requires source scoring sheets)
Adjective Rating	Good (as reported)

Acceptability	Acceptable (as reported)

Likert Scale and Computation of Results

The original five-point interpretation intervals are retained: 1.00–1.80 Strongly Disagree; 1.81–2.60 Disagree; 2.61–3.40 Undecided; 3.41–4.20 Agree; 4.21–5.00 Strongly Agree. The interval width is (5 − 1) / 5 = 0.80.

For one item, the frequency-weighted response mean is $M = \Sigma(f_i \times x_i) / N$, where f_i is the number of respondents choosing response value x_i, N is the number of valid responses, and Σ denotes summation across response categories. For a criterion with k equally weighted items, the average of item means is $WM = (\Sigma M_j) / k$, where M_j is the mean for item j. These quantities coincide only under the applicable weighting and response-completeness assumptions; the final manuscript should state which computation was actually used.

Reported Quality Evaluation Results

Criterion	Reported mean	Interpretation
Security	4.90	Strongly Agree
Performance efficiency	4.69	Strongly Agree
Reliability	4.88	Strongly Agree
Overall	4.82	Strongly Agree

Respondents strongly agreed with statements about security (4.90), performance efficiency (4.69), and reliability (4.88). The overall reported mean is 4.82. These are perception-based survey ratings; they do not independently prove absence of vulnerabilities, fast response times, or error-free operation. Technical claims must be supported by test records and measurements. Functional suitability was included in the Chapter 1 evaluation scope but lacks a separate reported respondent score here; attach actual functional-suitability evidence or align the scope and evaluation criteria across chapters.

[FIGURE 17 — replace the original weighted-mean screenshot with the verified final item-level table and recalculated criterion means. The source figure should not be treated as a validated calculation sheet.]

Appendices and Evidence Required Before Final Submission

• Final screenshots replacing outdated Figures 3–17; renumber any newly inserted figures and update in-text references.

• Original SUS questionnaire, anonymized response-level scoring sheet, item totals, and overall calculation.

• Likert instrument, per-item frequency tables, response totals, means and category aggregation worksheet.

• Biometric enrollment/trial log, valid and failed captures, timing measurements, and FAR/FRR calculation.

• Functional, security, payroll, offline, synchronization and duplicate-prevention test logs with observed outcomes.

• Deployment screenshots and configuration evidence for supported features and company rate classifications.